



THESIS ASSIGNMENT

Name and Surname: Jakub Novotný
Study programme: Data Science (Joint degree study, bachelor I. deg., full time form)
Field of Study: Computer Science
Mathematics
Type of Thesis: Bachelor's thesis
Language of Thesis: English
Secondary language: Slovak

Title: Machine Learning Assessment of the Trading Value of Central Bank Information

Annotation: The thesis will explore how financial markets incorporate central bank information and how this information affects the value of asset portfolios in high-frequency settings. Using high-frequency equity prices and derivatives data (such as swaps and futures), the student will attempt to measure the market's reaction to central bank communications, data releases, and monetary policy surprises. The empirical task will involve designing and implementing a machine learning model that learns from past market reactions to infer the trading value of central bank information. The model will dynamically allocate investments across a portfolio of assets, adjusting in real time based on observed return patterns and volatility responses with aim to maximize returns under historical and simulated conditions.
Beyond its academic contribution, the thesis serves as a preparatory stage for an NBS research project on the informational value of central bank communications. If successful, the results may feed directly into policy-relevant research and lead to a co-authored paper with NBS researchers.

Supervisor: Ing. Pavel Gertler, PhD.
Department: FMFI.KI - Department of Computer Science
Head of department: prof. RNDr. Martin Škoviera, PhD.

Assigned: 30.10.2025

Approved: 04.11.2025 doc. Mgr. Tomáš Vinař, PhD.
Guarantor of Study Programme

Student

Supervisor